

Vehicle Security via Federated Learning

Building a Collaborative Intrusion Detection System (IDS)

The Nervous System of the Car

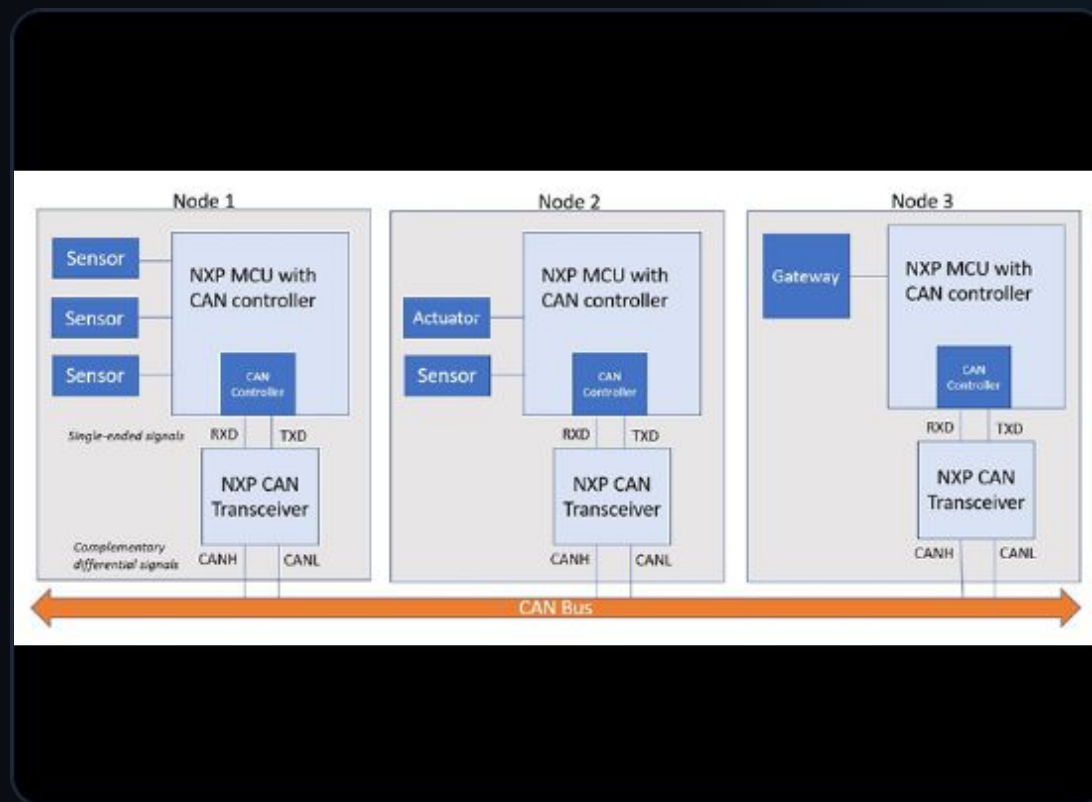
Analyzing the Controller Area Network (CAN) Framework

Vulnerabilities of the CAN Bus

Legacy by Design: Developed in 1983 for closed, isolated environments where external threats were not anticipated.

- **No Authentication:** Nodes cannot verify the identity of the message sender.
- **Shared Bus:** Every ECU sees every message, simplifying eavesdropping.
- **No Encryption:** Messages are sent in plaintext, allowing for easy spoofing.

Impact: Attackers can inject malicious frames via OBD-II ports or remote interfaces.



The Threat Landscape



Flooding

Overwhelming the bus with high-frequency messages to jam legitimate communications.



Fuzzing

Injecting random data payloads to discover unexpected or dangerous ECU behaviors.



Replay

Capturing valid traffic and re-transmitting it at incorrect times to deceive safety systems.



Suppression

Blocking critical signals (like braking or engine RPM) from reaching their destination.



Masquerade

The hardest to detect: carefully crafted messages that contain subtly incorrect data.



Zero-Day

Novel attack vectors that haven't been previously identified or labeled.

Collaborative Defense

Harnessing Unsupervised Intelligence & Federated Learning

Supervised Classification

Models are trained on **labeled** examples of both "normal" and "attack" traffic.

The Advantage

Delivers exceptionally high accuracy on **known** attack types because the model has directly memorized specific malicious signatures.

The Critical Failures

-  **Data Bottleneck:** Requires labeled labels. OEMs do not intentionally attack their own cars to generate training data.
-  **Zero-Day Blindness:** Completely fails on new attack types it has never seen before.

The Case for Unsupervised Models



No Labeled Data

Real vehicles rarely encounter labeled attacks. Unsupervised models train exclusively on "normal" operational data observed during routine driving.



Anomaly Detection

By learning to reconstruct normal signals, the model identifies attacks as high-error outliers. This enables the detection of unknown threats.



Scalability

Unsupervised Autoencoders are lightweight and fit the constrained memory profiles of modern ECUs while maintaining high detection recall.

Unsupervised Anomaly Detection

Models learn from **normal traffic only**, using reconstruction error to identify outliers.

The Advantage

- ✓ **Real-World Ready:** No labeled attack data required for deployment.
- 🛡️ **Catch-All:** Can identify zero-day attacks by detecting deviations from normal patterns.

The Challenge

The Masquerade Threat: Subtle attacks (like slightly off RPM values) might look normal enough that the reconstruction error doesn't spike, allowing them to slip past.

The Verdict: Why Unsupervised?



Operational Viability

Automakers train on what they observe during routine driving. Unsupervised is the only path that mirrors real-world deployment.



Federated Synergy

In Federated Learning, local training is limited to individual vehicle observations. Only unsupervised models can learn in this environment.



Resource Efficiency

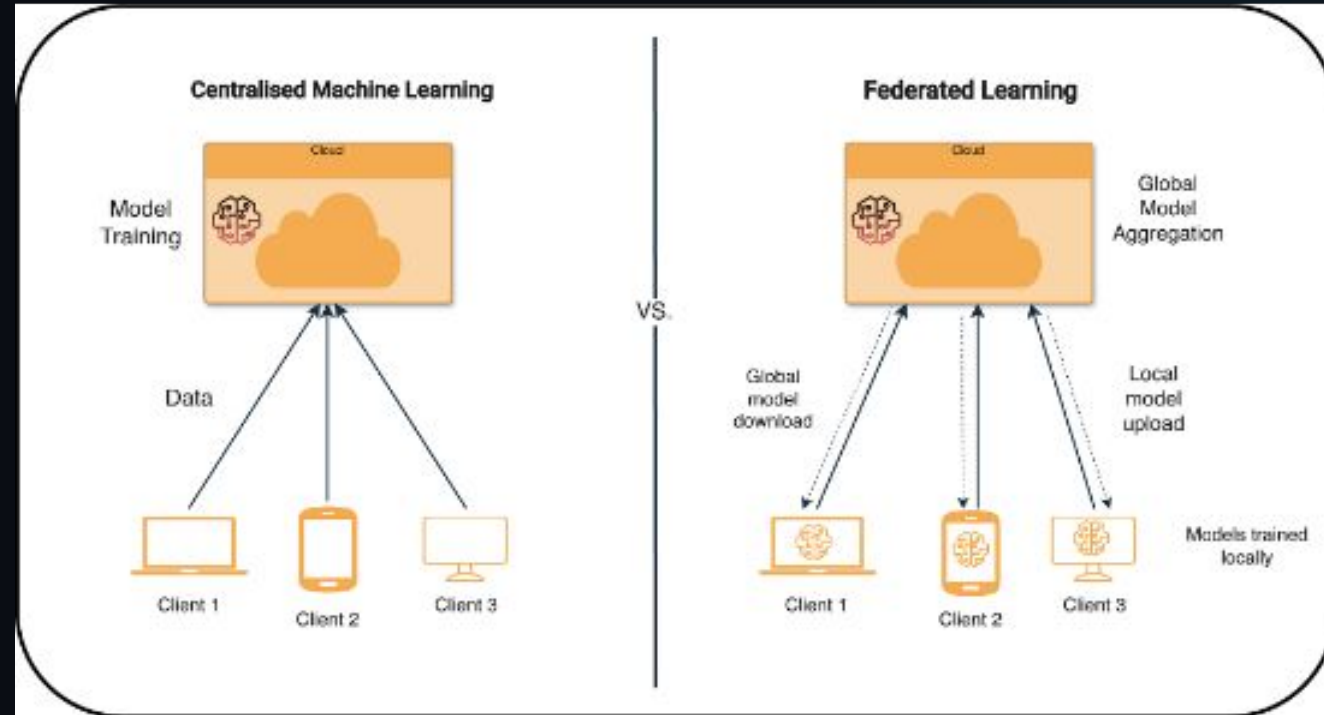
Autoencoders learn compressed representations, keeping the model lightweight for Electronic Control Units (ECUs).

Selected Approach: Unsupervised Autoencoder (CANet Architecture)

Federated Learning Framework

Privacy-First Collaboration: Vehicles improve their local IDS models by sharing mathematical updates instead of raw sensor data.

- **Local Training:** Each car trains on its unique CAN bus traffic patterns.
- **Global Aggregation:** A server combines updates to build a robust, fleet-wide intelligence.
- **Continuous Loop:** The global model is redistributed to improve local safety.



Deliverable 1: Threat Model

Category	Definition & Scope
System Protected	The internal CAN bus of a connected vehicle within a Federated network.
Adversary Profile	In-vehicle attacker with CAN access (Remote/Physical) injecting malicious frames.
Adversary Limits	Attacker cannot modify the IDS model or access detection thresholds.
Trust Model	Nodes are honest-but-curious; local data never leaves the vehicle.

Strategic Detection Goals

0.90+

TARGET AUC

0.85+

F1-SCORE

Detection metrics must ensure **High Recall** (missing an attack is critical)
and **Precision** (preventing driver alarm fatigue).

Deliverable 2:

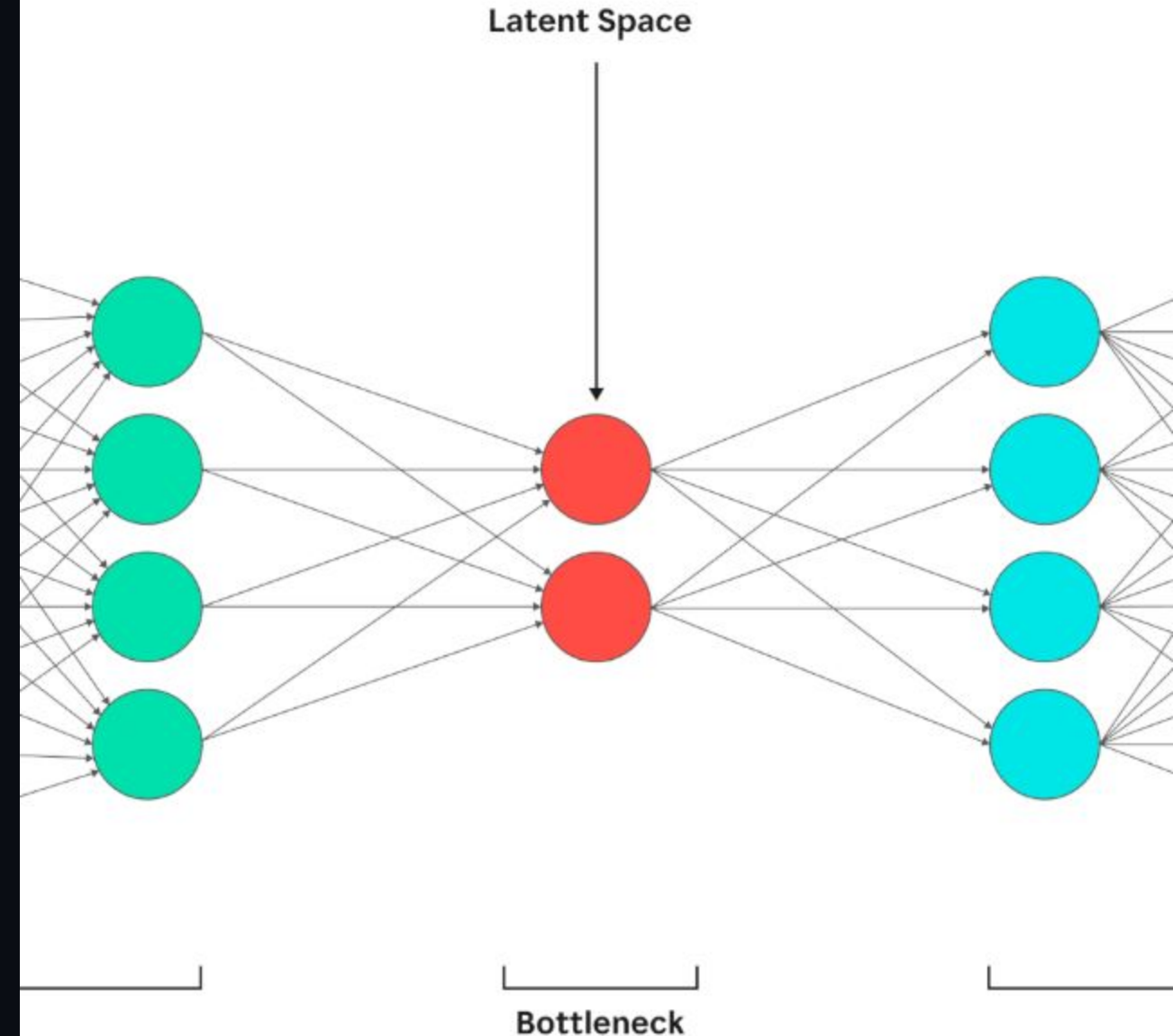
Baseline Architecture

The **CANet Architecture**: A deep autoencoder designed to minimize reconstruction error on normal traffic.

⚡ **Encoder**: Compresses high-dimensional CAN signals into a latent representation.

🔲 **Decoder**: Reconstructs the signals from the compressed latent space.

📈 **Loss**: Mean Squared Error (MSE) used to measure deviations.



Dataset Evaluation Pipeline

Phase 1: SynCAN

Synthetic data used for initial pipeline setup. Clean, structured, and labeled numerical signals.

Phase 2: Car-Hacking

Raw Hyundai hex IDs and payloads. Real-world challenge for data parsing and hex-decoding.

Phase 3: X-CANIDS

Production-level complexity. Deserialized signal representations using DBC database definitions.

Implementation Roadmap



Environment: Python 3.10/3.11 with PyTorch.



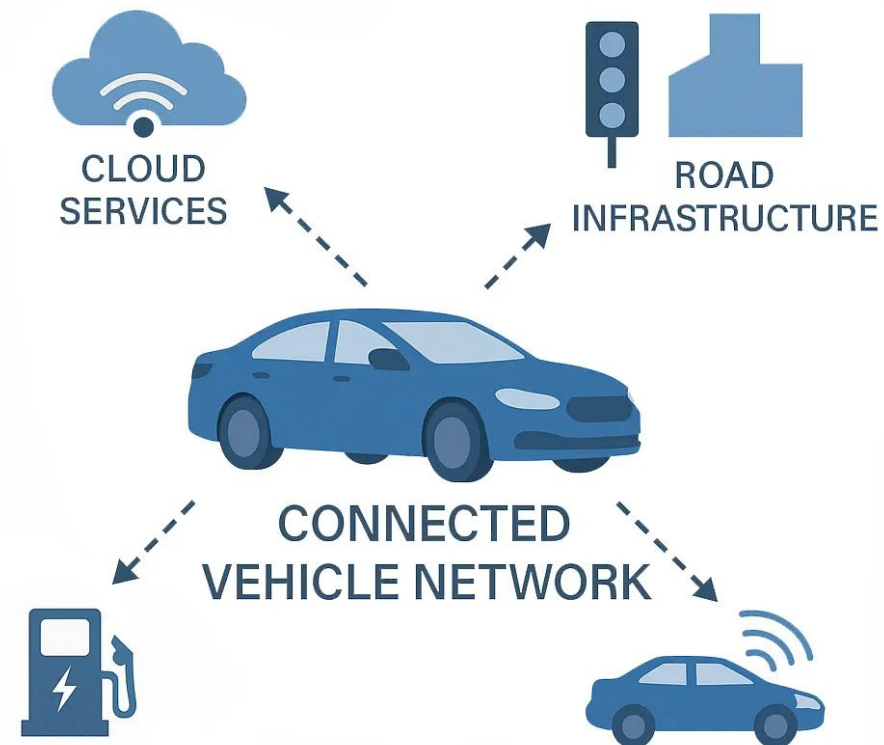
Libraries: Pandas, NumPy, Scikit-learn, Seaborn.



Parameter Count: Required to establish baseline for fleet scaling.



Documentation: Finalizing the Threat Model & Architecture docs.



Q&A

Ensuring Robustness in Vehicular Intrusion Detection

Next Step: Calculating Parameter $\text{numel}()$ for Global Scaling

Image Sources



<https://community.nxp.com/t5/image/serverpage/image-id/135116i51F5A3B8663D1BC1?v=v2>

Source: community.nxp.com



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